

Supporting Pre-Release Decision Making in Indian Cinema

A Data-Driven Analysis of Certification, Content, and Release Timing on Opening Weekend Box Office Performance

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Register No. CB.SC.U4CSE23146 | Class/Section: CSE-B

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Abstract. Producers, distributors, and multiplex chains in India commit to costly pre-release decisions — certification category, release timing, and scale of theatrical distribution — weeks before a film’s actual audience response is known. This case study builds a self-scraped dataset of 1,900 Indian films across nine language industries and the years 2013–2026 to identify which pre-release attributes drive opening-weekend box office collection, and to test whether the effect of CBFC certification (U/UA/A) on that collection has shifted between the pre-COVID and post-COVID periods. Two ordinary least squares models (standard errors clustered by film, since pan-India releases contribute one row per language version) and a Random Forest predictive model are estimated. Star power, runtime, sequel status, and festival-window timing are robust, statistically significant drivers of opening weekend. The central finding is that certification’s effect has genuinely reversed: adult (A) certified films, the weakest-opening category before the pandemic, are the strongest-opening category afterward, while U-certified films have fallen the furthest (interaction coefficient -0.667 , $p = 0.004$). This holds across four successive dataset expansions and within Hindi alone. The predictive model explains 60% of the variance in log opening weekend on a film-grouped test split. Findings are compared against three recent published studies and translated into recommendations for certification strategy, release timing, and screen allocation.

1. Problem Statement and Objectives

Producers, distributors, and multiplex chains in India commit to costly, high-stakes decisions — certification category (U/UA/A), release timing, and scale of theatrical distribution — weeks before a film’s actual audience response is known, often relying on intuition and star power rather than evidence from historical performance. This creates real financial risk: an unnecessary editing decision may be made to secure a family-friendly certificate, or a distributor may over-commit screens and marketing spend to a film that does not warrant it. This is important because screens, release slots, and marketing budgets are scarce, expensive resources, and post-pandemic shifts in theatrical habits and OTT adoption have made pre-2020 intuition an increasingly unreliable guide [4].

This study analyses how pre-release attributes — genre, language, certification, runtime, cast track record, sequel status, and release timing — relate to a film’s opening-weekend box-office collection, and specifically examines whether the influence of certification has shifted between the pre-COVID and post-COVID periods. The objectives are:

1. Identify which pre-release factors significantly influence opening-weekend box-office performance.

2. Build a predictive model that estimates an expected opening-weekend collection range from attributes known before release.
3. Determine whether the effect of certification on opening-weekend performance has changed between the pre-COVID (2013–2019) and post-COVID (2022–2026) periods, and derive decision-support recommendations for producers, distributors, and multiplex chains.

2. Data Collection and Dataset Description

2.1 Initial Approach and Its Limits

Data collection began by compiling candidate film titles from Wikipedia’s yearly film-list pages, one per language per year (for example, “List of Hindi films of 2023” or “List of Tamil films of 2019”), for five Indian language industries (Hindi, Tamil, Telugu, Malayalam, Kannada) — chosen because all five share the same Central Board of Film Certification (CBFC) system, keeping certification directly comparable across languages, which an international sample would not allow. Each candidate title was matched to a page on Sacnilk (sacnilk.com), a publicly accessible Indian box-office data platform, by guessing the page’s URL slug from the title. This approach succeeded roughly 15% of the time and, after three rounds of increasingly aggressive candidate sourcing, plateaued at 396 films.

2.2 Discovery of Sacnilk’s Internal Records API

Inspecting the network traffic behind Sacnilk’s own box-office-records pages revealed that those pages are populated by a request to an internal JSON endpoint that returns, for each film, an exact page reference, release date, and a pre-computed opening-weekend figure — correctly computed per language for multi-language releases (verified directly against a known pan-India release, where the API’s per-language figure matched an independent day-wise reconstruction to two decimal places). Querying this endpoint directly, once per language per year, eliminated slug-guessing entirely: of 715 films identified as new through this method, zero failed to resolve to a real page, against the roughly 85% failure rate of the guessing approach. This single change took the dataset from 396 to 1,111 films and is the primary reason the final dataset substantially exceeds the scale originally proposed.

2.3 Expansion Beyond the Query Cap

A further expansion round probed the API’s limits directly rather than assuming them. Each query returns at most 50 films with no working pagination; additional films in dense buckets were recovered by also querying the API’s day-1, week-1, month-1, and cumulative-collection-club rankings. Hindi-language data exists back to 2013, and four further industries — Punjabi, Gujarati, Marathi, and Bengali — are present but only from 2022 onward. Most consequentially, *regional-language data before 2019 is genuinely sparse in Sacnilk’s own system*: querying Tamil, Telugu, Malayalam, or Kannada for 2016 or 2017 returns zero or near-zero films, a property of the underlying data source rather than a limitation of this study’s scraping method. This round added 789 further films (86 were subsequently excluded for an incomplete opening-weekend record; see Section 3.1), bringing the final dataset to 1,900 films.

2.4 Auxiliary Sources

Wikipedia film infoboxes and article text were scraped to fill residual gaps in runtime and certification. The Movie Database (TMDb) public API was used, via its documented endpoints, to obtain India-specific certification (as a cross-check), audience ratings and vote counts, budget, and cast popularity — all bundled into a single request per film using TMDb’s `append_to_response` parameter.

2.5 Final Dataset Composition

The final dataset contains 1,900 films spanning 2013–2026, divided into three periods: pre-COVID (2013–2019, 391 films), during-COVID (2020–2021, 170 films, analysed separately since theatrical release patterns were disrupted rather than simply different), and post-COVID (2022–2026, 1,339 films). Table 1 shows the composition by language and period. Ninety percent of pre-COVID films (350 of 391) are Hindi; four industries have no pre-2022 representation at all. This constraint shapes the interpretation of every pre-/post-COVID comparison in this study and is addressed through a Hindi-only robustness check (Section 6.2).

Table 1: Dataset Composition by Language and Period

Language	Pre-COVID	2020–21	Post-COVID	Cert. %
Hindi	350	32	257	84%
Telugu	17	51	276	72%
Tamil	15	29	250	78%
Malayalam	6	18	203	85%
Kannada	3	20	113	49%
Punjabi	0	11	79	29%
Marathi	0	3	68	39%
Gujarati	0	1	49	80%
Bengali	0	5	44	73%
Total	391	170	1,339	73.8%

Fifteen attributes are used in the analysis: title, year, language, period, opening-weekend collection in Rs. Crore (the outcome variable), genre, certification (U/UA/A), runtime, release date, festival-window flag, sequel flag, a same-dataset recurring-cast flag, a time-aware cast-track-record measure (Section 4.1), TMDb rating, and budget. Coverage: genre 98.3%, rating 95.3%, runtime 83.5%, certification 73.8%, and (real, placeholder-corrected) budget 45.9%.

3. Data Preparation and Exploratory Analysis

3.1 Data Cleaning and Quality Corrections

Five data-quality issues were identified and corrected during the study.

Wrong-language duplication. A small number of titles were found scraped under two different language labels but resolving to the identical underlying Sacnilk page with an identical opening-weekend figure — the signature of a single-language film mistakenly matched under a second language via an incidental Wikipedia title collision. These were removed (3 films). Genuine simultaneous multi-language releases were verified to report a distinct figure per language and were retained as separate, non-independent observations, handled via clustering (Section 4.2).

Incomplete opening-weekend records. 34 films for which a full Friday–Sunday day-wise collection window could not be reconstructed were excluded rather than retained with a silently understated figure.

Rating placeholder. Sacnilk’s own “user rating” field was found to display a fixed placeholder value for any film with no genuine audience engagement rather than real data. It was replaced throughout with TMDb’s audience rating.

Budget placeholder. Sacnilk records an unknown budget as Rs. 1 Crore rather than leaving the field blank. An earlier version of this analysis reported budget coverage of 76.8% and a budget–revenue correlation of 0.50 without detecting this; on inspection, 472 of 872 “budgets” in that version were exactly Rs. 1 Crore. With placeholders excluded, real budget coverage is 45.9% and the log-scale

correlation with opening weekend is 0.70 (Section 3.5). This correction is reported explicitly, since silently overstated coverage in an earlier draft could otherwise mislead a reader who consulted only the final figures.

Certification standardisation. CBFC’s 2024 reform introduced age-banded UA sub-ratings (UA 7+/13+/16+) alongside the pre-existing U/UA/A scheme. All UA sub-bands were collapsed into UA so that pre- and post-COVID certification values remain directly comparable.

3.2 Target Variable and Temporal Structure

Opening-weekend collection is heavily right-skewed (Fig. 1): most films earn modestly while a small number of blockbusters dominate the scale. All modelling below uses $\log(1 + x)$ of this variable.

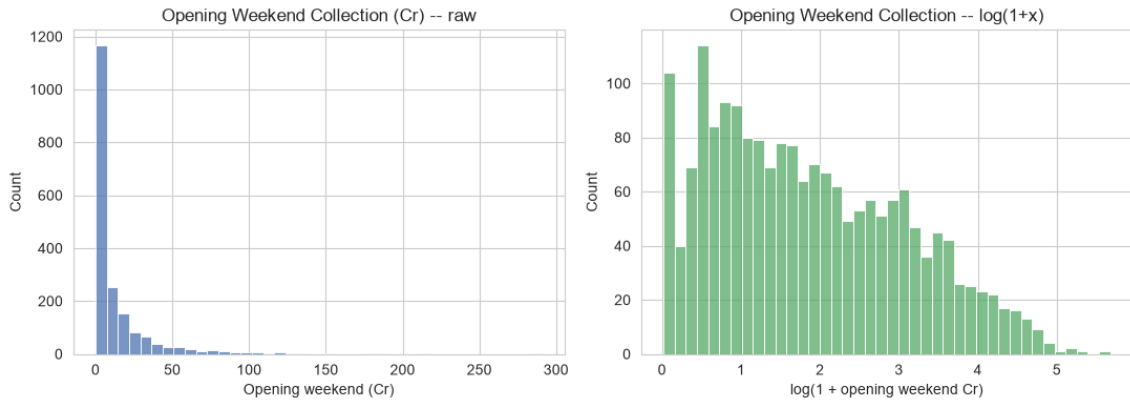


Figure 1: Distribution of opening-weekend collection, raw (left) and log-transformed (right). The long right tail motivates the log transform used in all regression and prediction models.

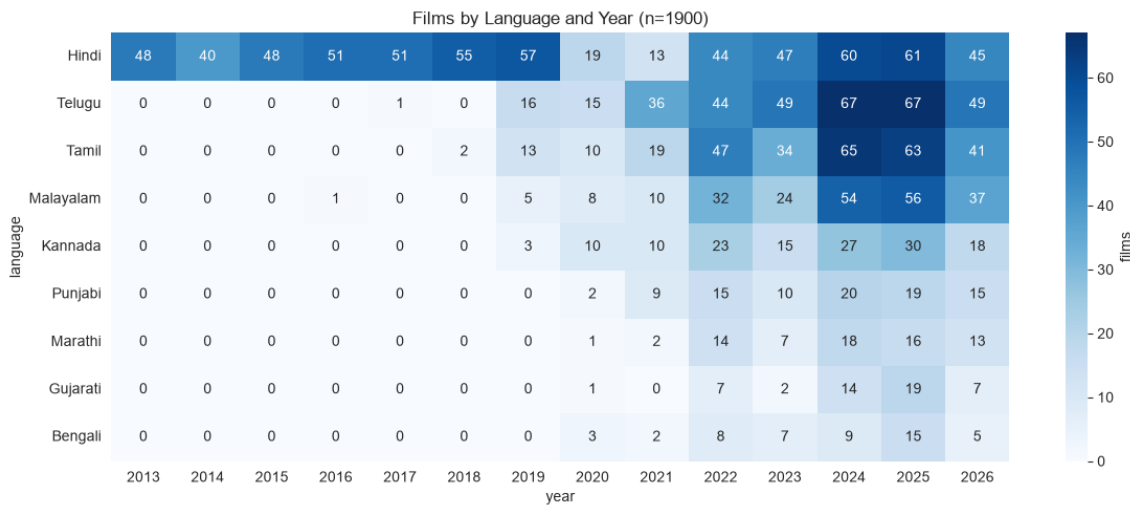


Figure 2: Number of films by language and year. Hindi is the only industry with continuous pre-COVID coverage; four industries have no representation before 2022. This asymmetry, verified directly against Sacnilk’s own historical records, is a property of the data source and motivates the Hindi-only robustness check in Section 6.2.

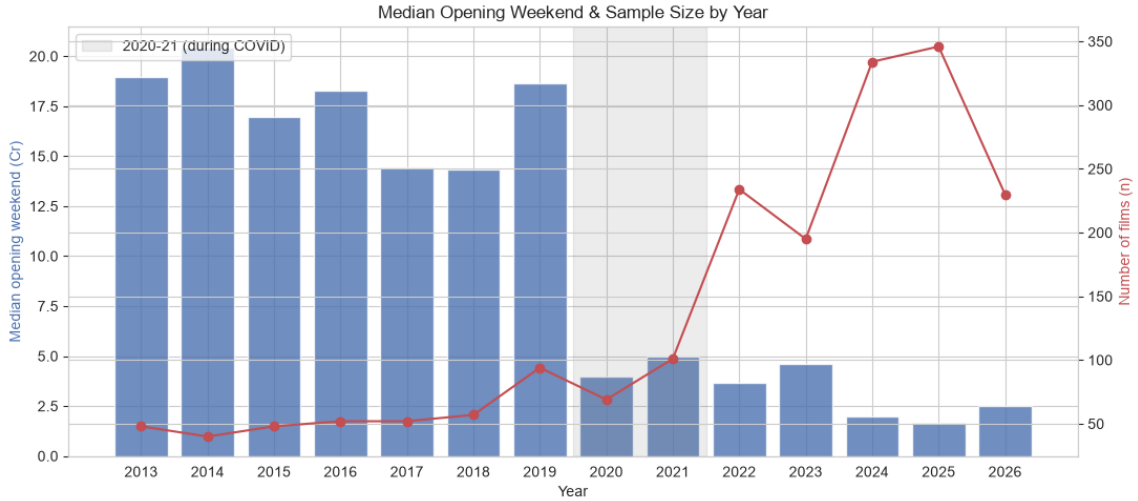


Figure 3: Median opening-weekend collection and sample size by year, 2013–2026. A step down is visible around 2020–2021 and persists afterward even as coverage grows substantially in later years.

Fig. 3 shows the year-wise median falling from roughly Rs. 14–20 Crore before COVID to Rs. 1.7–4.6 Crore after — consistent with, though not on its own proof of, a structural change in opening-weekend behaviour, since the post-COVID years also contain far more small-budget and regional releases (Fig. 2).

3.3 Period and Certification Effects

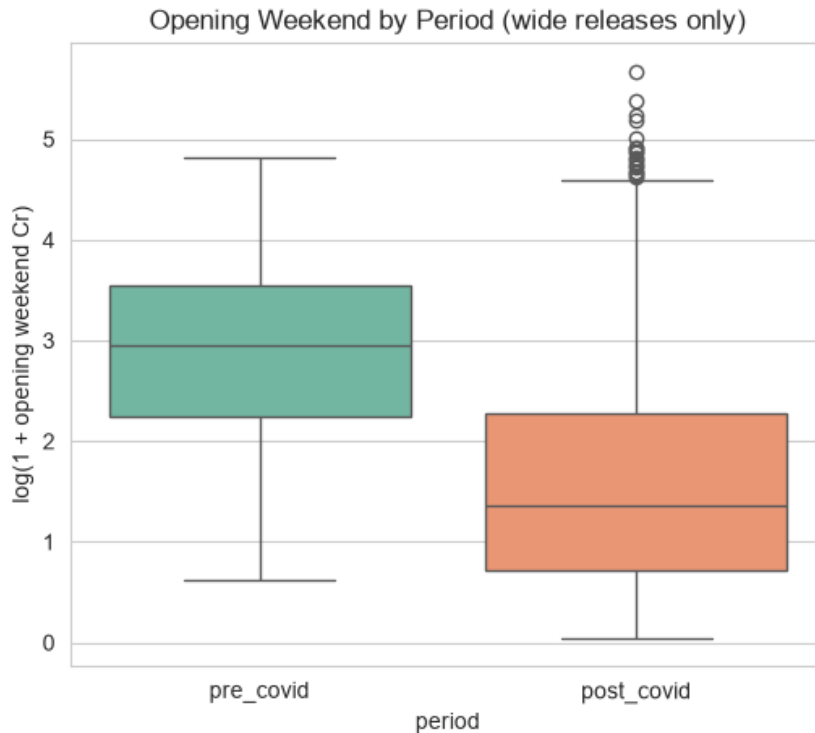


Figure 4: Opening weekend by period, wide-release films only. Pre-COVID films open substantially higher; part of this gap reflects language composition and is decomposed in Section 6.1.

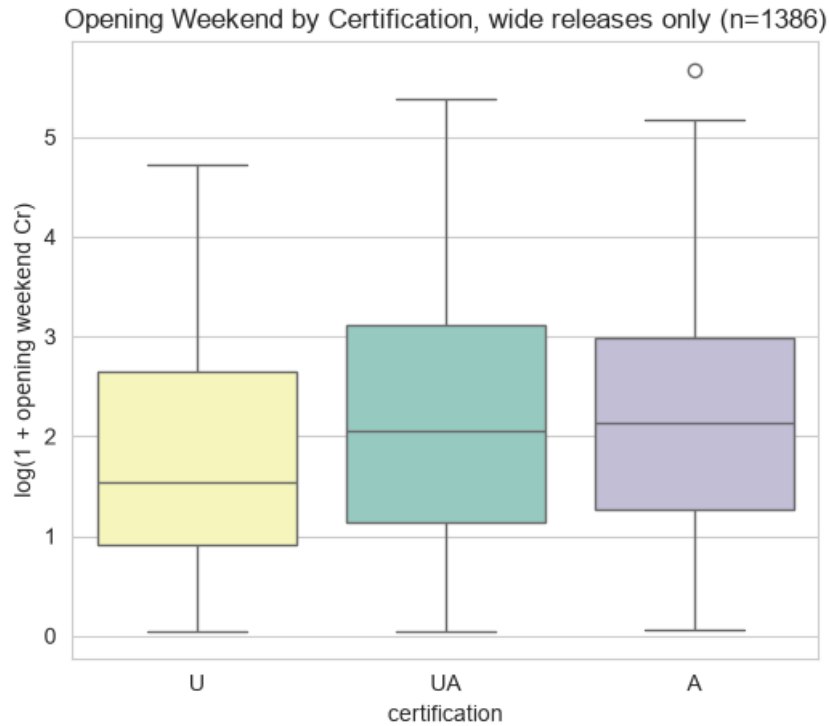


Figure 5: Opening weekend by certification category, wide-release films only, pooled across both periods. UA and A films open above U films on average.

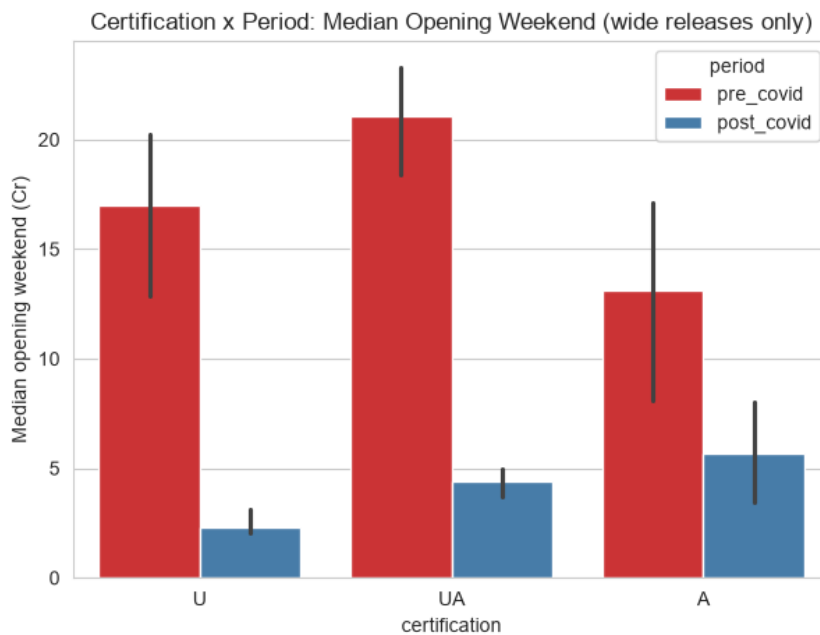


Figure 6: Certification $\times$ period: median opening weekend by CBFC category and period. The category ordering reverses: A-certified films move from the weakest-opening to the strongest-opening category; U-certified films fall the furthest. Formally tested in Section 6.2 (Model B, $p = 0.004$).

Fig. 6 is the central exploratory result of this study. Before COVID, A-certified films had the lowest median opening weekend of the three categories; after COVID, they have the highest, while U-certified films — the highest-opening category pre-COVID — have fallen the furthest.

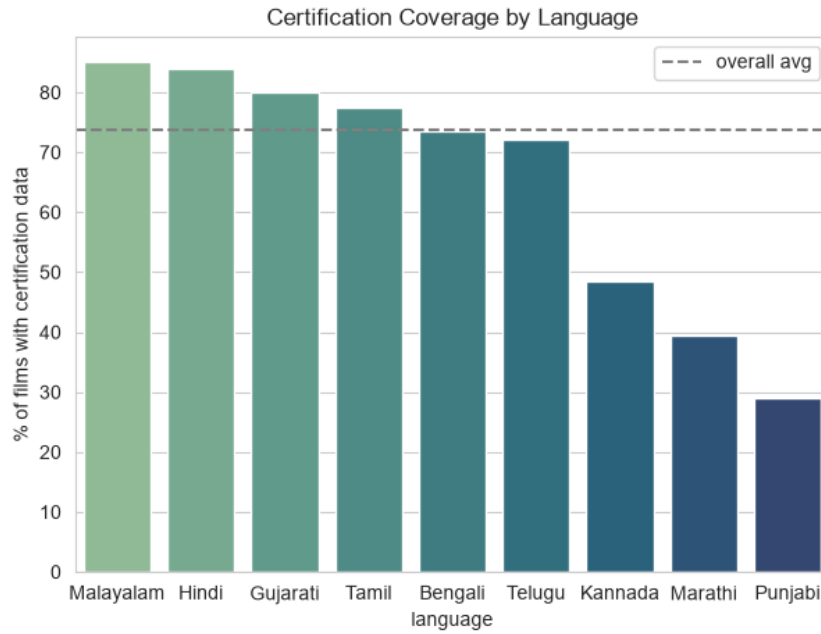


Figure 7: Certification coverage by language. Malayalam, Hindi, Gujarati, Tamil, Bengali, and Telugu are well covered (72–85%); Kannada, Marathi, and Punjabi are not (29–49%), and certification findings for those three languages should be treated with corresponding caution.

3.4 Language and Content Effects

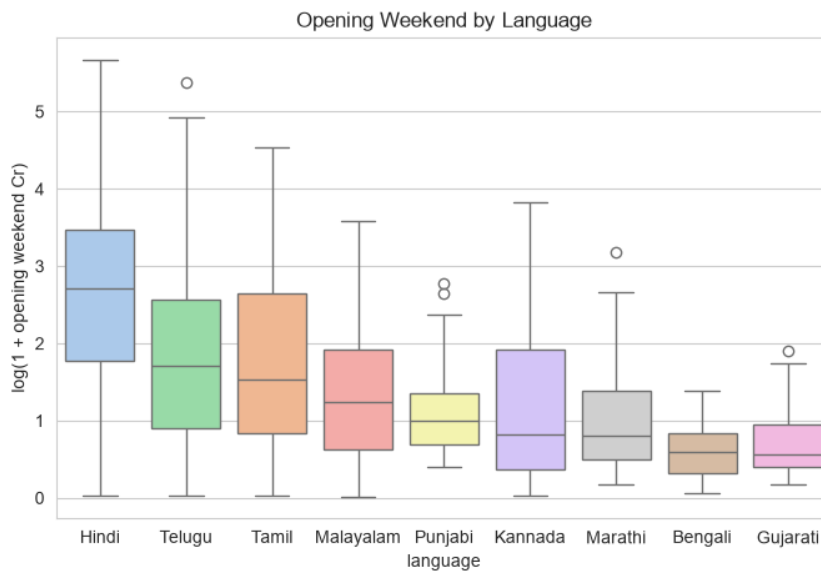


Figure 8: Opening weekend by language. Hindi opens clearly above every other industry.

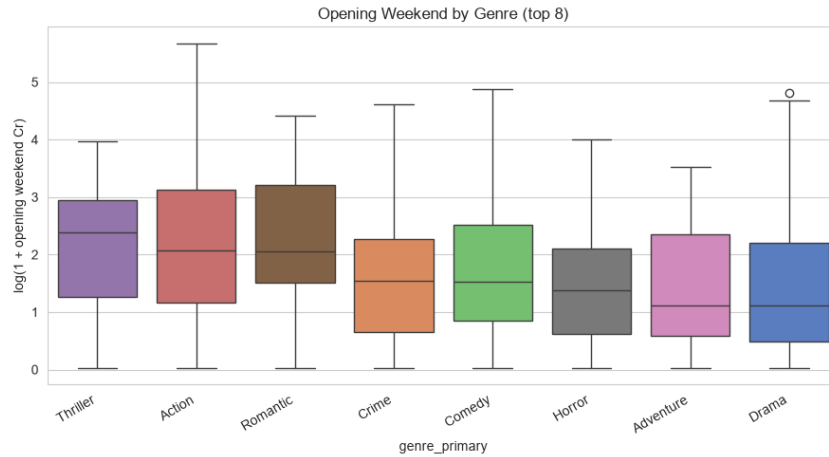


Figure 9: Opening weekend by genre (top 8 by frequency). Thriller’s apparently high raw median is driven by a small number of large Hindi thrillers ($n=41$) and reverses once language and other factors are controlled for (Section 6.1).



Figure 10: Opening weekend: festival-window release vs. other releases. Festival releases open substantially higher and this effect survives regression control (Section 6.1, $p = 0.002$), unlike on an earlier, smaller version of this dataset where only 21 festival films were available to test it.

3.5 Cast, Rating, and Budget Signals

“Star power” has no single agreed definition, so three candidate measures were built and compared directly against opening weekend, rather than assuming one upfront.

Same-dataset recurring-cast flag (a binary indicator that some member of the cast appears in at least one other film in this dataset). Fig. 11 shows a large raw gap between flagged and unflagged films. However, as the dataset expanded to 1,900 films, this flag has grown to mark 69% of all films, making it a considerably blunter instrument than it was on earlier, smaller versions of this dataset.

TMDb live popularity score (the cast’s current popularity metric, queried at analysis time). Fig. 12 shows this correlates only weakly with opening weekend (0.19). The likely reason is a timing mismatch: this score reflects an actor’s fame *today*, which can differ substantially from their standing at the time a given film, possibly years old, actually opened.

Time-aware cast track record (introduced in Section 4.1): for each film’s top-3 billed cast, the mean log opening weekend of that actor’s *earlier* films only, taking the best of the three. This measure correlates far more strongly with opening weekend (0.62; Fig. 13) and, unlike the other two, is free of look-ahead bias — it uses only information that would genuinely have been available before the film’s own release.

Because it is both the strongest predictor and the only one that respects release-time information, the time-aware measure is adopted as the primary cast-quality feature in the predictive model (Sections 4.1 and 6.3), while the simpler recurring-cast flag is retained in the explanatory models (Section 4.2) for continuity with the dataset’s earlier development stages and is itself revisited as a robustness check in Section 6.2.

Two further pre-release-adjacent signals were checked for completeness, though neither is used as a model feature. TMDb audience rating shows essentially no relationship with opening weekend (correlation 0.09; Fig. 16) — unsurprising, since a rating is formed only after audiences have seen the film, while opening weekend is set by expectations formed beforehand. Budget, once Sacnilk’s Rs. 1 Crore placeholder values are excluded (Section 3.1), correlates moderately with opening weekend (0.31 raw, 0.70 log-scale; Fig. 15), but is left out of the models because it covers under half the dataset and overlaps substantially with cast track record and release scale, which are measured separately and with better coverage.

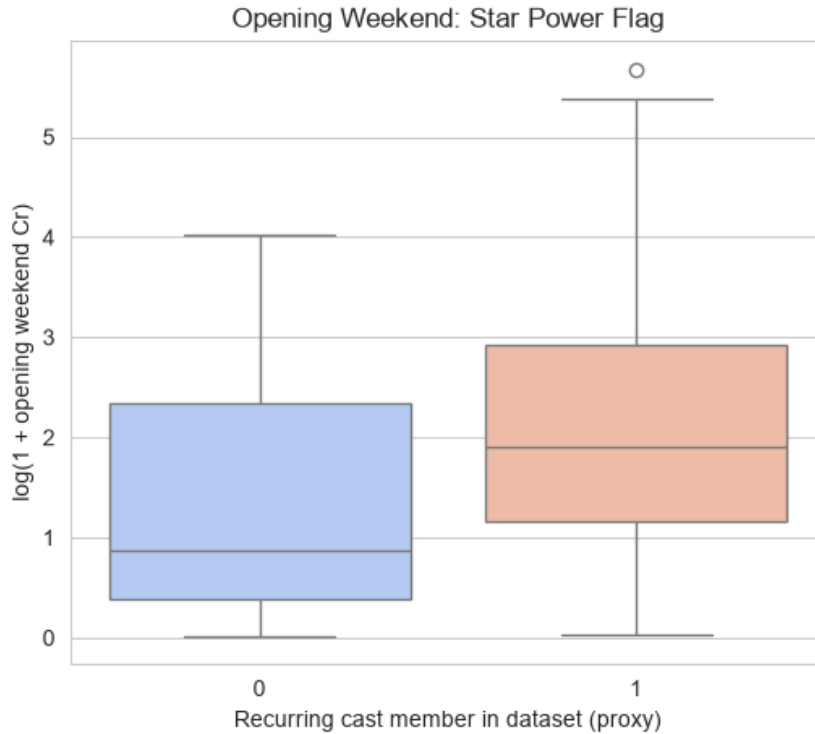


Figure 11: Opening weekend by same-dataset recurring-cast flag. A large gap, but the flag has become less discriminating as the dataset grew (now marks 69% of films).

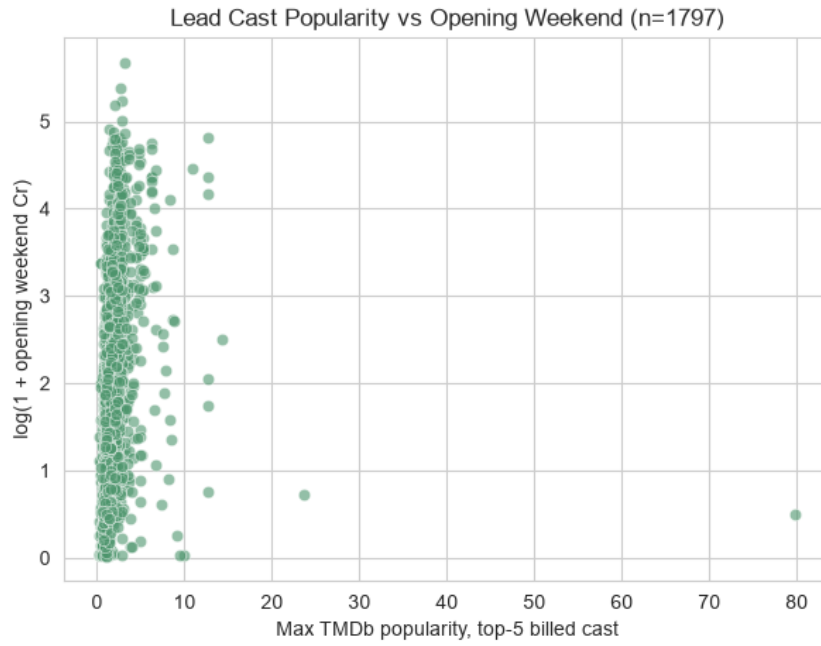


Figure 12: TMDb live popularity score vs. opening weekend (correlation 0.19). Weak, likely because the score reflects present-day fame rather than fame at release.

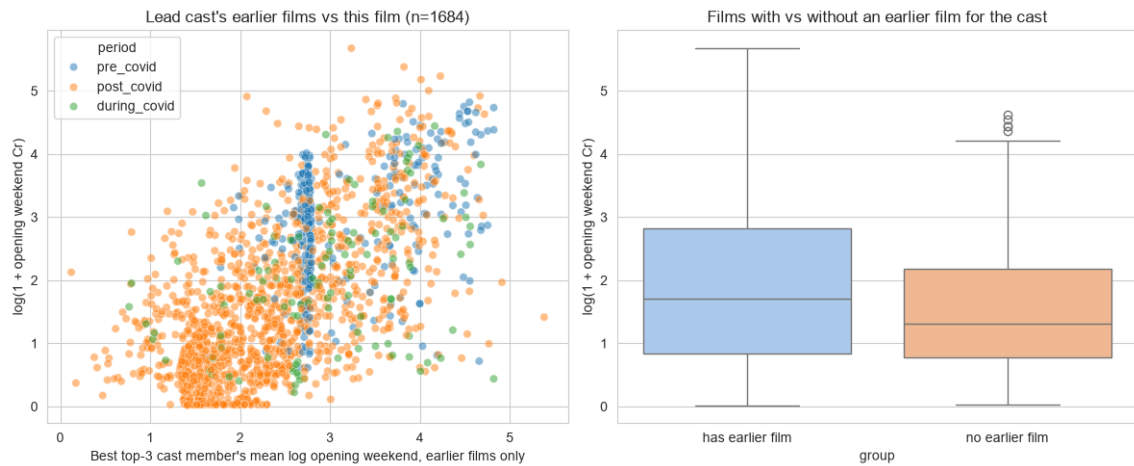


Figure 13: Time-aware cast track record vs. opening weekend (left; correlation 0.62) and films with vs. without an earlier film for any top-3 cast member (right; 11% of films have none, and open noticeably smaller). This measure uses only information available before release and is the strongest single predictor identified in this study (Section 6.3).

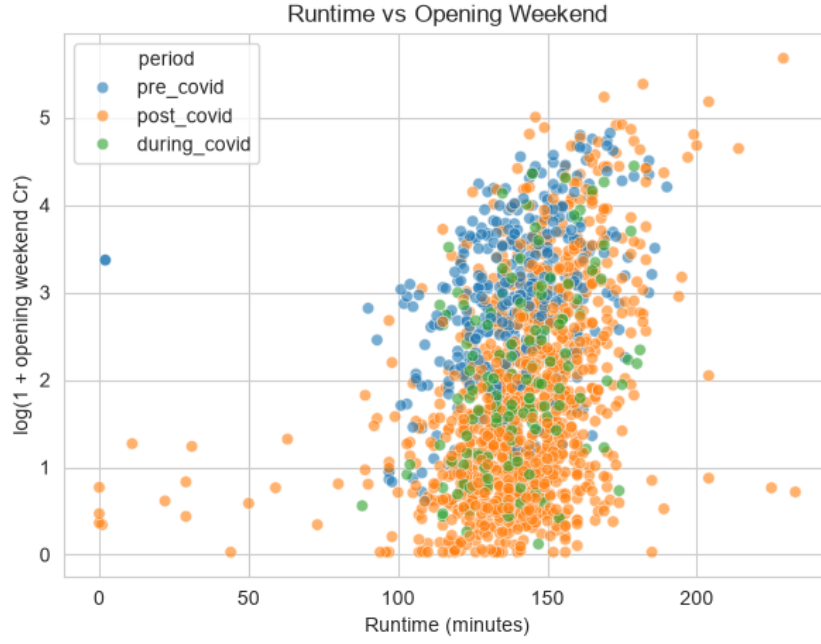


Figure 14: Runtime vs. opening weekend (correlation 0.35). A modest positive relationship that remains statistically significant in the full regression.

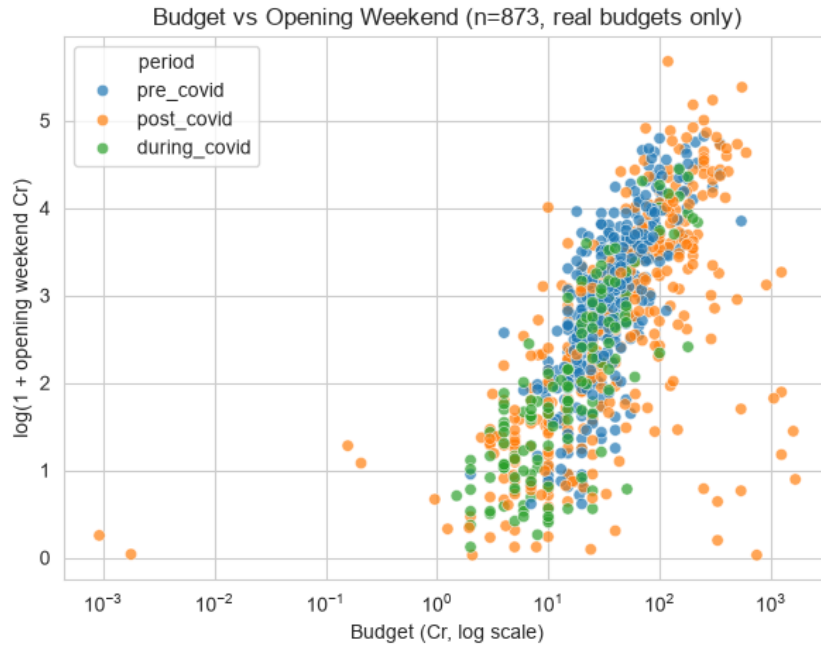


Figure 15: Budget vs. opening weekend, real budgets only (placeholder Rs. 1 Crore values excluded; $n = 873$). Correlation 0.31 on the raw scale, 0.70 on the log scale. Not used as a model feature since it covers under half the dataset and substantially overlaps with cast track record and release scale.

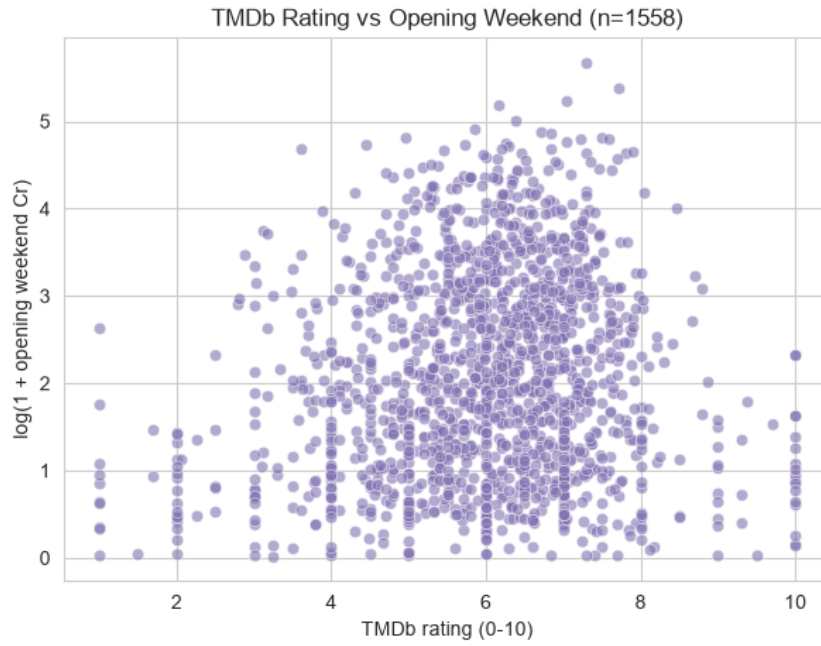


Figure 16: TMDb audience rating vs. opening weekend (correlation 0.09). Essentially no relationship: opening weekend is set by pre-release expectation, while the rating is formed only after audiences have seen the film.

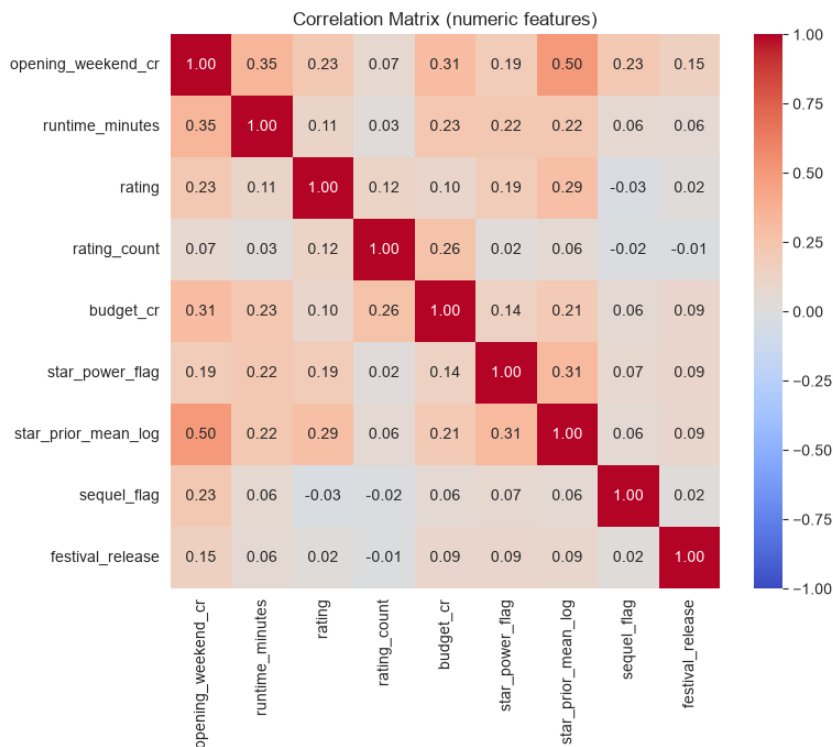


Figure 17: Correlation matrix of numeric features. No pair is dangerously collinear, so multicollinearity was not a concern in specifying the regression models of Section 4.

4. Analytics Method and Implementation

4.1 Feature Engineering

Two engineered features are introduced beyond the attributes directly scraped. A *sequel flag* is derived from the title (detecting numerals, ordinal words, and terms such as “Part”, “Returns”, or “Chapter”). The *time-aware cast-track-record* measure described in Section 3.5 is computed by processing films in release-date order and, for each film, looking backward only: for its top-3 billed cast, the mean log opening weekend of each actor’s strictly earlier films in the dataset is computed, and the maximum across the three actors is retained, together with a count of how many earlier films the best-tracked actor has. Films for which no top-3 cast member has an earlier film in the dataset (11.4%) are flagged separately rather than imputed with a misleading zero.

4.2 Explanatory Regression Models

Multiple linear regression (OLS) was selected for the explanatory analysis because Objective 3 requires a testable interaction term with an interpretable sign and magnitude, which a black-box model does not straightforwardly provide. Two specifications were estimated on $\log(1 + \text{opening weekend})$.

Model A regresses on genre, language, runtime, the recurring-cast flag, sequel status, festival-window timing, and period (pre-COVID / during-COVID / post-COVID, post-COVID as reference), using all films with genre and runtime recorded ($n = 1,555$), to address Objective 1.

Model B restricts to wide-release, certification-known films in the pre- or post-COVID period ($n = 1,261$; during-COVID excluded from this specific test) and adds a certification $\times$ period interaction term (Treatment coding, U and post-COVID as joint reference), while controlling for language, directly addressing Objective 3.

In both models, *standard errors are clustered by film*, because a pan-India release contributes one row per language version, and these rows share cast, budget, and genre — treating them as independent observations would understate standard errors and overstate statistical significance.

4.3 Predictive Model

A Random Forest regressor (300 trees, maximum depth 6, minimum 4 samples per leaf) was benchmarked against a linear regression baseline for Objective 2, using genre, language, certification (with an explicit “Unknown” category rather than row deletion), period, runtime (with a missing-value indicator and median imputation), the recurring-cast flag, festival timing, sequel status, and the time-aware cast-track-record measure. Critically, the train/test split (80/20) was performed *by film* rather than by row: since language versions of one film share nearly all of these attributes, a row-level random split would let the model implicitly recognise test-set films through their training-set language-version counterparts, inflating the reported accuracy. An 80% empirical prediction interval was constructed from the residual distribution of the held-out test set, so that a point prediction can be reported alongside a defensible range, as required by Objective 2.

5. Comparison with State-of-the-Art Methods

Box-office prediction and the economics of film certification have both been studied extensively, though rarely together and rarely for the Indian, multi-language theatrical market specifically. Verma and Verma [1] compared six supervised learning algorithms for predicting whether a Bollywood film would be a commercial hit or flop, using pre-release attributes including lead-actor rating, IMDb rating, music rating, and planned screen count; Random Forest and Logistic Regression achieved 92% classification accuracy. Paul and Das [2] addressed early-stage prediction of Bollywood films’ international revenue, motivated by the industry’s shift from sequential to simultaneous worldwide release — directly relevant

to this study’s treatment of pan-India multi-language releases. Lampe and McRae [3] examined 1,922 films across 31 countries (2002–2011), finding that restrictive age classifications reduce box-office revenue, particularly for films with wide commercial appeal; this is the closest prior work to Objective 3 and provides the benchmark against which the present study’s India-specific, COVID-era finding is compared below. None of these three studies analyses more than one Indian language industry, tests a certification×time interaction with clustered inference for multi-language releases, or validates a predictive model on a film-grouped train/test split — the specific gap this study addresses. Table 2 presents the required comparison.

Table 2: Comparison with State-of-the-Art Methods

Verma & Verma (2020) [1]	
Dataset	Bollywood films, hit/flop labelled
Method	DT, RF, SVM, Logistic Regression, AdaBoost, ANN (classification)
Eval. metric	Classification accuracy
Key result	RF and Logistic Regression: 92% accuracy; IMDb rating among top predictors
Comparison with this work	Models opening weekend as a continuous, log-transformed outcome across 9 languages rather than a Hindi-only binary label. Notably, rating was a top predictor there but shows almost no relationship with opening weekend here ($r = 0.09$); the two are not contradictory since rating is post-release information, while opening weekend is a pre-release expectation outcome.
Paul & Das (2023) [2]	
Dataset	Bollywood films, international revenue
Method	Multiple ML algorithms; models simultaneous vs. sequential multi-market release
Eval. metric	Improvement over baseline
Key result	Early-stage features meaningfully improve revenue prediction
Comparison with this work	Both predict revenue from pre-release attributes for Bollywood; this work is domestic-only but spans 9 languages and treats simultaneous multi-language release as repeated, non-independent observations of one film via clustered inference, rather than as a release-strategy feature.
Lampe & McRae (2021) [3]	
Dataset	1,922 films, 31 countries, 2002–2011
Method	Panel regression, self- vs. state-regulated ratings
Eval. metric	Effect size on revenue by rating stringency
Key result	Restrictive ratings reduce revenue, especially for wide-appeal films
Comparison with this work	Both test whether certification affects revenue. Lampe & McRae find a consistently negative effect of restrictive ratings in a pre-pandemic, cross-country window; this work finds that sign has reversed in India specifically after COVID. The results are compatible, not contradictory: they describe different market/time contexts and jointly suggest the certification–revenue relationship is not fixed but shifts with audience composition.

None of the three studies is treated as superior or inferior on the basis of a single reported score, since datasets, countries, and time windows differ substantially. The comparison instead highlights where this study’s findings agree with, extend, or usefully complicate the published literature — most notably, the certification-effect reversal, which is a genuinely new observation relative to Lampe and McRae’s earlier, pre-pandemic finding.

6. Results, Business Insights and Recommendations

6.1 Model A: General Drivers of Opening Weekend

Table 3 reports Model A ($n = 1,555$, $R^2 = 0.519$, clustered by film). Star power (recurring-cast flag), sequel status, and runtime are the strongest, cleanest drivers, all significant at $p < 0.001$. Pre-COVID

films open significantly higher even after controlling for language, genre, runtime, and cast (coefficient +0.981), confirming that Fig. 4’s raw gap is not purely a language-composition artefact. All eight non-Hindi languages open significantly below Hindi, and several genres (Drama, Mystery, Comedy, Thriller, Biography, Crime) open significantly below Action.

Table 3: Model A: Drivers of Opening Weekend ($n = 1,555$, $R^2 = 0.519$)

Variable	Coef.	p -value
Period: pre-COVID (vs. post)	+0.981	< 0.001
Recurring-cast flag	+0.697	< 0.001
Sequel	+0.589	< 0.001
Runtime (per minute)	+0.017	< 0.001
Festival-window release	+0.233	0.002
Period: 2020–21 (vs. post)	+0.170	0.026
Language: Marathi (vs. Hindi)	−1.070	< 0.001
Language: Bengali	−1.028	< 0.001
Language: Kannada	−0.787	< 0.001
Language: Malayalam	−0.718	< 0.001
Language: Tamil	−0.430	< 0.001
Language: Telugu	−0.413	< 0.001
Genre: Mystery (vs. Action)	−0.527	< 0.001
Genre: Drama	−0.470	< 0.001
Genre: Thriller	−0.413	0.011
Genre: Comedy	−0.260	< 0.001

An R^2 of 0.52 for a cross-sectional, observational model of box-office revenue is consistent with, and toward the higher end of, values typically reported in this literature. Opening weekend is shaped by many factors this dataset cannot observe pre-release — marketing spend, trailer reception, competing releases in the same week, word-of-mouth, even weather — so a substantially higher R^2 from pre-release attributes alone would itself be a reason for suspicion rather than confidence. As a check, release month was added as a further control (not shown in Table 3): R^2 moved only from 0.519 to 0.528, suggesting the model is already close to the practical ceiling achievable from pre-release attributes on this dataset.

6.2 Model B: The Certification $\times$ Period Interaction

Table 4 reports Model B ($n = 1,261$, $R^2 = 0.310$, clustered by film). Post-COVID, both UA- and A-certified films open above U-certified films. The interaction term for A is negative and significant (-0.667 , $p = 0.004$): combined with the main effect, this implies that pre-COVID, A-certified films opened *below* U-certified films (net $0.291 - 0.667 = -0.376$), a reversal consistent with the raw medians in Fig. 6 (pre-COVID: A Rs. 13.15 Cr < U Rs. 17.01 Cr < UA Rs. 21.09 Cr; post-COVID: U Rs. 2.33 Cr < UA Rs. 4.40 Cr < A Rs. 5.65 Cr). The UA interaction is not significant ($p = 0.451$), so the reversal is specific to the A category.

Table 4: Model B: Certification $\times$ Period ($n = 1,261$, $R^2 = 0.310$)

Variable	Coef.	p -value
UA vs. U (post-COVID)	+0.245	0.005
A vs. U (post-COVID)	+0.291	0.066
Pre-COVID vs. post-COVID (for U)	+0.782	< 0.001
A $\times$ pre-COVID	−0.667	0.004
UA $\times$ pre-COVID	−0.116	0.451

Robustness — Hindi-only replication. Since pre-COVID data is 90% Hindi, the interaction was re-estimated within Hindi alone ($n = 507$). The reversal not only survives but sharpens: $A \times \text{pre-COVID} =$

-1.404 ($p < 0.001$), $UA \times \text{pre-COVID} = -0.845$ ($p = 0.007$) — within Hindi, both A and UA gained on U after COVID, confirming the finding is not a language-composition artefact.

Robustness — time-aware star power. Replacing the recurring-cast flag with the time-aware track-record measure in Model A’s specification improves fit ($R^2 = 0.540$ vs. 0.519), and the new measure is highly significant ($+0.449$, $p < 0.001$), confirming it is not merely a theoretically cleaner alternative but an empirically better one.

Robustness — sensitivity to additional controls. Model B is deliberately kept lean (certification, period, and language only) to isolate the interaction with maximum power in a subsample already restricted by certification availability. As a transparency check, general box-office drivers from Model A were added one group at a time; Table 5 reports the results. R^2 rises substantially (0.310 to 0.506), but the $A \times \text{pre-COVID}$ interaction weakens and loses significance at conventional levels once these controls are present.

Table 5: Sensitivity of the Certification $\times$ Period Interaction to Added Controls

Specification	R^2	$A \times \text{pre coef.}$	$p\text{-value}$
Lean (primary; Table 4)	0.310	-0.667	0.004
+ Runtime	0.425	-0.330	0.132
+ Runtime + Genre	0.453	-0.256	0.245
+ Runtime + Genre + Cast flag + Sequel	0.506	-0.182	0.398

Most of this attenuation traces to one variable: the recurring-cast flag is itself mechanically tied to period (85% of post-COVID films are flagged vs. only 52% pre-COVID, simply because the larger post-COVID sample yields more repeat-cast detections — a dataset-growth artefact, not a real casting shift; Section 3.5). Controlling for it therefore partly absorbs the effect it should be independent of. Runtime and genre, being pre-treatment attributes, attenuate the interaction more moderately (rows 2–3, vs. row 4). The finding is best read as a robust association in its primary, intended specification and within Hindi alone — not a fully control-proof causal estimate — and Section 6.4’s recommendations are worded accordingly.

Stability across dataset versions. Table 6 reports the interaction across four successive, independently-collected dataset versions. The effect was not significant on the first, thinnest sample (only two pre-COVID A-certified films existed then) and has been significant, in the same direction, on every subsequent expansion, with the point estimate stabilising toward a more moderate magnitude as sample size grew — the expected signature of an initially noisy small-sample estimate being corrected by more data.

Table 6: Stability of the Certification $\times$ Period Finding

Dataset stage	n	Pre-COVID A	$A \times \text{pre (p)}$
Wikipedia-candidate	203	2	-1.15 (0.185, n.s.)
+ Cert. backfill	260	5	-1.67 (0.009)
+ Records-API exp.	822	13	-0.88 (0.033)
Final (clustered)	1,261	38	-0.67 (0.004)

6.3 Predictive Model

The Random Forest model modestly outperforms the linear baseline on a film-grouped hold-out set: $R^2 = 0.602$, $\text{MAE} \approx \text{Rs. } 8.07$ Cr, versus $R^2 = 0.569$, $\text{MAE} \approx \text{Rs. } 8.40$ Cr for linear regression. The modest gain indicates the underlying relationship is mostly additive rather than sharply non-linear. Fig. 18 shows feature importances: the time-aware cast-track-record measure dominates (0.54), ahead of runtime (0.15) and the count of the lead cast’s earlier films (0.14); certification barely registers,

because a marginal importance score cannot see a conditional (interaction) effect of the kind Model B is specifically built to detect.

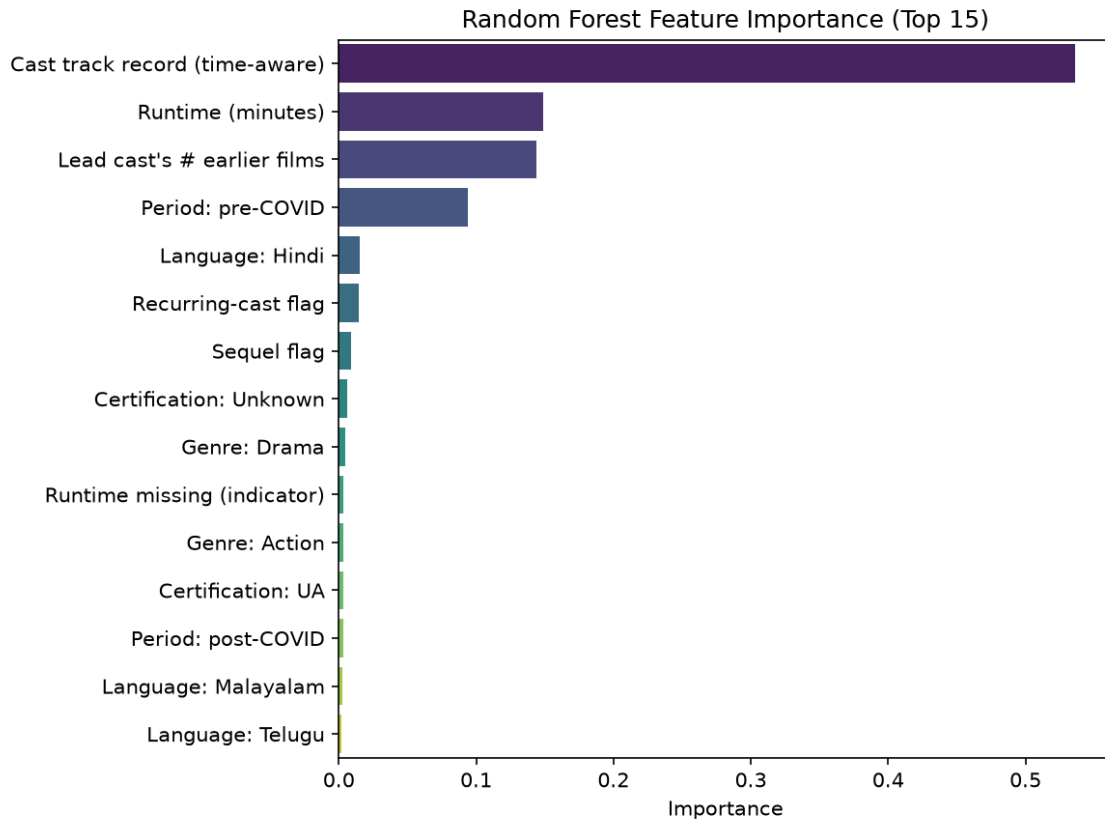


Figure 18: Random Forest feature importance (top 15). The time-aware cast-track-record measure dominates, consistent with its strong performance as a standalone predictor (Fig. 13) and its significance in the Model A robustness check.

For a representative example — a UA-certified Hindi action film, 140 minutes, not a sequel, with a cast at a typical (median) track-record level, released post-Covid outside a festival window — the model's point estimate is Rs. 4.36 Cr, with an 80% empirical prediction interval of Rs. 1.12–12.25 Cr. This width is itself an operationally important finding: pre-release attributes narrow expectations but do not eliminate genuine box-office uncertainty.

6.4 Business Insights and Recommendations

Certification strategy (producers). The historical assumption that a lenient (U) certificate is the commercially safer choice is not supported by post-2022 data: U-certified films now have the lowest median opening weekend of the three categories and have fallen furthest since the pandemic. An edit made purely to secure a U certificate should not be assumed to reduce commercial risk. This is not, conversely, a case for deliberately pursuing an A certificate: the effect describes how films with each certificate *have* opened, not what would happen if a specific film were re-cut to a different rating. This recommendation should also be read with the caveat established in Section 6.2: the interaction is robust and significant in its primary, focused specification and within Hindi alone, but attenuates once general content controls (genre, runtime) are added, meaning part of the observed shift may reflect broader changes in what kind of films get made and certified, not certification in isolation.

Release timing (distributors). Festival-window releases open approximately 26% higher ($p = 0.002$) after controlling for genre, language, runtime, cast, and sequel status. Screen and marketing commit-

ments timed to Pongal/Sankranti, Holi, Eid, Independence Day, Diwali, and Christmas windows are supported by the data.

Screen allocation and marketing sizing (producers/distributors). A lead cast’s track record in their own earlier films is the single strongest pre-release signal identified (Fig. 18) — more informative than a simple “does the film have a star” flag — and should be weighted accordingly. Sequels can be expected to open roughly $1.8\times$ higher than a comparable non-sequel.

Expectation-setting (multiplex chains). The 80% prediction interval for a representative mid-tier film (Rs. 1.12–12.25 Cr around a Rs. 4.36 Cr point estimate) implies that fixed, single-number screen commitments carry real risk; flexible allocation with room to adjust after the first day is the more defensible operating approach.

Market-specific benchmarking. All eight non-Hindi industries open significantly below Hindi even after controlling for other factors (Table 3), most plausibly reflecting market scale rather than content quality; distributors should benchmark expectations by language market rather than against a single national figure.

7. Conclusion and References

Limitations

Pre-COVID language imbalance. Ninety percent of pre-COVID films are Hindi, because Sacnilk’s own historical tracking of the other eight industries barely extends before 2019–2020 — verified directly by probing the API across every year and ranking type, and therefore a property of the data source rather than of the scraping method. Every pre-/post-COVID conclusion in this study was consequently cross-checked within Hindi alone.

Uneven certification coverage. Certification coverage is 73.8% overall but only 29–49% for Kannada, Marathi, and Punjabi; certification-based conclusions describe the certified subsample and should not be extended to those three languages specifically.

Star-power measurement. None of the three cast-quality measures tested is perfect: the same-dataset flag has become blunt as the dataset grew (marks 69% of films); TMDb popularity is measured at query time, not at release; and the time-aware measure, while empirically the strongest, is undefined for 11.4% of films whose top-3 cast has no earlier film within this dataset.

Sample composition. A large share of films entered the dataset via Sacnilk’s own ranked “top opening weekend” lists, which likely skew toward more notable theatrical releases; the dataset represents tracked releases, not a uniform random sample of every film that opened in a cinema.

Correlational, not causal, inference. All results derive from observational data. The certification $\times$ period finding is a robust, replicated statistical association, not evidence that any individual film’s certification decision caused its box-office outcome.

Conclusion

This study analysed 1,900 Indian films across nine languages and the years 2013–2026 to identify the drivers of opening-weekend box-office performance and to test whether certification’s effect on that performance has shifted since COVID-19. Star power, runtime, sequel status, and festival-window timing are robust, statistically significant drivers of opening weekend. The central finding is that the effect of CBFC certification on the ordering of certification categories has reversed: adult (A) certified films, the weakest-opening category before the pandemic, are the strongest-opening category afterward, while family-friendly (U) certified films have fallen the furthest. This result held up under four successive, independent rounds of dataset expansion and a same-language (Hindi-only) robustness check, and it runs

counter to the pre-pandemic, cross-country evidence of Lampe and McRae [3], suggesting that India’s post-pandemic theatrical audience composition has shifted in a way that favours adult-certified content. This finding is robust in its primary, focused specification, but — reported transparently in Section 6.2 — attenuates once general content controls such as genre and runtime are added, so it should be understood as a genuine, replicated statistical association rather than a fully control-proof causal claim. A Random Forest model built on these attributes explains 60% of the variance in log opening weekend on a rigorously film-grouped test split, sufficient for range-based planning though not for precise revenue forecasting, given the intrinsic unpredictability of individual blockbuster outcomes.

Future work could extend this analysis in three directions: incorporating marketing-spend and social-media pre-release buzz data, where publicly available; modelling the certification $\times$ period interaction jointly with OTT release-window length, which itself changed substantially over the study period [4]; and, as regional-language historical data becomes more thoroughly digitised by sources such as Sacnilk, revisiting the pre-COVID comparison for Tamil, Telugu, Malayalam, and Kannada with a sample size comparable to Hindi’s.

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